

Classification of waste materials in very-high-resolution satellite imagery

State of the art and thesis proposal

Alessandro Potenza · M.Sc. Computer Science and Engineering, AI · PERIVALLON

Context

Illegal waste dumping is an environmental crime with direct public-health consequences.

Environmental agencies (ARPA) have limited inspection capacity. Intervention priority depends on what is dumped: inert rubble, plastics and asbestos-cement imply very different hazards.

Automatic detection of dump sites from aerial and satellite images is mature. Recognising the material is not: the reference survey lists 50 works from 1987 to 2023, almost all RGB, and marks material identification as an open problem.

Within PERIVALLON, one thesis in the group has addressed material classification directly (Alari 2024). This work starts from there.

Task definition

Task: multi-label classification of waste materials in satellite images, at image level.

Input: very-high-resolution optical imagery, GSD 0.2 to 1.3 m: aerial RGB for the baseline (AerialWaste), satellite VNIR for the multispectral arm. Bands: RGB and VNIR (up to 8). SWIR is excluded: not part of the planned acquisitions, and its 3.7 m GSD does not match the sub-metre task. Sentinel-2 (10-20 m) is excluded: too coarse.

Output: the set of materials present in each image, from a fixed list of categories.

Technique: classification, not object detection or segmentation. The available annotations are image-level (Alari 2024: over 11,400 multi-label annotations). Waste piles have no stable shape for boxes, and no segmentation masks exist for these datasets.

Research question: does VNIR multispectral input improve waste-material classification compared with RGB, and for which materials?

Materials: which are considered and why

Material	Classification target	Spectral analysis (RGB vs VNIR)	Note
Rubble / inert	yes	yes	frequent; grey, confusable in RGB
Plastic	yes	yes	same colour as other bright surfaces in RGB
Wood and firewood	yes	yes	confusable with soil and vegetation
Tires	yes	yes	dark target, low contrast
Asbestos-cement	yes	yes, dedicated pilot	public regional ground truth (WFS)
Vehicles, tanks, containers	yes	no	shape-based, RGB sufficient in literature
Scrap, bulky items, big bags	yes	no	shape-based, few spectral cues expected
Sludge, foundry waste	yes	no	few labels; visually ambiguous at this GSD

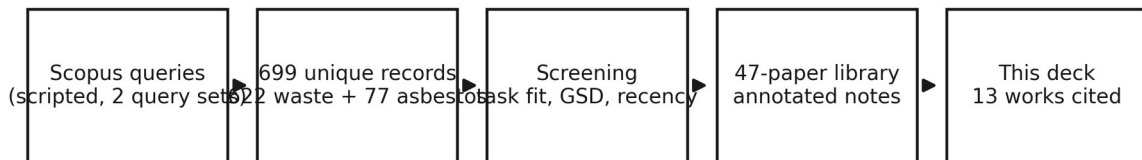
Literature search: method and numbers

Scripted queries on the Scopus API, two query sets: waste detection in remote sensing; asbestos roof mapping.

After deduplication: 699 unique records (622 waste, 77 asbestos).

Screening criteria: task pertinence (terrestrial waste or roof materials), GSD compatible with the task, recency and citations, peer review. Snowballing from the Fraternali 2024 survey and from the Alari 2024 thesis references.

Result: a curated library of 47 papers with structured notes, synchronised with the team Excel. 13 works are cited in this deck.



Related work: site-level waste detection

Work (year)	Input / GSD	Task	Method	Result
Gibellini 2025	aerial RGB, 20 cm	classification	Swin-T, RSP pretraining	F1 92.0; cross-region -5.1
Disaitek 2024	Pléiades Neo, 30 cm	detection	operational service	~95% on sites >2 m2 (vendor)
CascadeDumpNet 2024	Pléiades, 50 cm	object detection	two-stage CNN + AutoML	mAP 84.6, cross-city transfer
Sun 2023	VHR RGB, 0.3-1 m	object detection	BCA-Net (Faster R-CNN)	~2,500 dumpsites, 28 cities
CWLD 2024	GF-2 + GE, 0.5-0.8 m	segmentation	DeepLabV3+ variant	F1 88.9, construction waste
AerialWaste, Torres 2023	aerial RGB, 20-50 cm	dataset + cls.	ResNet-FPN baseline	F1 80.7; 22 material tags

Related work: material-level classification

Work (year)	Input / GSD	Task	Method	Result
Alari 2024 (PoliMi)	satellite RGB	multi-label cls.	ResNet / Swin + FPN	wF1 69.2 (5 cat.), 59.4 (10 cat.)
Saba 2026	WV-3 VNIR, 1.24 m	pixel cls.	Fine-KNN (32 tested)	asbestos, Macro-F1 97.6
Bonifazi 2026	WV-3 VNIR+SWIR, 1.24/3.7 m	pixel cls.	max likelihood, temporal	asbestos roofs, removal tracking
Abbasi 2024	aerial RGB	OBIA cls.	DenseNet + LSTM	asbestos, OA ~96 (shape + time)
Cilia 2015	airborne HSI, 3 m	pixel cls.	SAM + weathering index	asbestos, PA 89 / UA 86

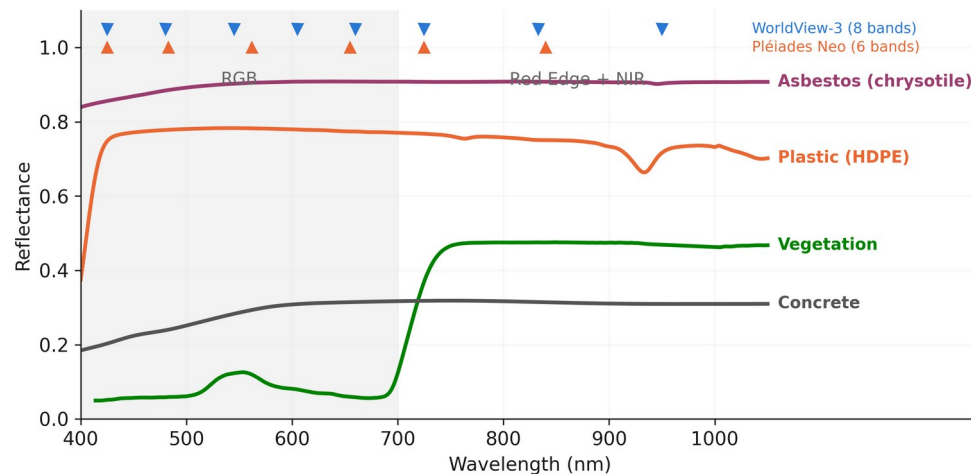
Where RGB falls short, and what VNIR adds

Different materials share the same colour at 0.3-1.3 m: plastic sheets, asbestos-cement and concrete all appear grey.

In the group predecessor, moving from 5 to 10 material categories costs 9.8 F1 points (69.2 to 59.4). Finer material distinctions are the hard part.

Red Edge and NIR separate vegetation, bare soil and weathered surfaces; in Saba 2026 they drive asbestos discrimination.

Whether this helps waste materials, and which ones, has not been measured. It is the object of this thesis.



Available imagery

	WorldView-3	Pléiades Neo
Multispectral	8 VNIR bands, 1.24 m	6 VNIR bands, 1.2 m
Panchromatic	0.31 m	0.30 m
Extra bands vs RGB+NIR	Coastal, Yellow, Red Edge, NIR2	Deep Blue, Red Edge
Access	commercial / ESA proposal	commercial / ESA proposal

Both sensors are sub-metre with pan-sharpening and carry VNIR beyond RGB. This is the band budget of the study: no SWIR, no Sentinel-2.

What is missing in the literature

1. Multi-material classification has one direct precedent, with ample margin: wF1 59-69 (Alari 2024).
2. No work measures the added value of VNIR bands over RGB for waste materials at very high resolution.
3. Results are reported as aggregate scores; per-material behaviour is not analysed.
4. Generalisation across regions is rarely evaluated. At site level it costs 5 F1 points (Gibellini 2025).
5. Asbestos is studied on roofs, in isolation, and never inside a waste-material taxonomy.

Proposed work: approach

Technique: multi-label image classification, continuing the group line (Gibellini 2025, Alari 2024).

Backbone: Swin-T with remote-sensing pretraining, the current group baseline. Extra VNIR bands enter by extending the input layer of the pretrained network.

Data: AerialWaste with the Alari material annotations as RGB baseline; WorldView-3 / Pléiades Neo acquisitions for the multispectral arm; Lombardy WFS asbestos registry (10,903 roofs) as ground truth for a controlled pilot.

Band ablation: RGB, then RGB + NIR, then full VNIR. Same architecture, same splits, only the input changes.

Asbestos pilot first: public labels, single material, clear VNIR evidence in literature. It validates the pipeline before the full taxonomy.

Proposed work: evaluation

Per-material F1 alongside weighted and macro averages. Aggregates hide exactly the classes this thesis targets.

Delta versus the RGB baseline for each band configuration, with confidence intervals over repeated runs.

Generalisation protocol: train and test on disjoint geographic areas, in addition to the standard split.

Reference points: wF1 69.2 / 59.4 (Alari 2024) for the multi-label task; Macro-F1 97.6 (Saba 2026, per-pixel) as upper reference for the asbestos pilot, not directly comparable.

Outcome either way: if VNIR does not help a material, that is a documented negative result with practical value for sensor choice.

References

- Alari 2024. Fighting environmental crime with deep learning: classifying waste materials from illegal landfills in satellite imagery. M.Sc. thesis, PoliMi, 10589/230633.
- Fraternali et al. 2024. Solid waste detection, monitoring and mapping in remote sensing images: a survey. arXiv:2402.09066.
- Gibellini et al. 2025. A deep learning pipeline for solid waste detection in remote sensing images. Waste Management Bulletin.
- Torres, Fraternali 2023. AerialWaste: a dataset for illegal landfill discovery in aerial images. Scientific Data.
- Zhang, Ma 2024. CascadeDumpNet. Remote Sensing of Environment. · Sun et al. 2023. Nature Communications 14. · CWLD 2024. Scientific Data.
- Saba et al. 2026. J. Hazardous Materials. · Bonifazi et al. 2026. Geomatics. · Abbasi et al. 2024. RSASE. · Cilia et al. 2015. ISPRS IJGI.
- Kokaly et al. 2017. USGS Spectral Library v7 (splib07a). USGS Data Series 1035. · Disaitek 2024, vendor report (Airbus).